

- 7 (a) (i) Since the 4 branches have the same p.d., they are in parallel arrangement.

$$\frac{1}{R} = \frac{1}{20} + \frac{1}{40} + \frac{1}{20} + \frac{1}{40}$$

$$R = 6.66 \, \Omega$$

- (ii) Consider power delivered to each of the lamps,

$$P = \frac{V^2}{R}$$

$$\text{For M and P, } P = \frac{E^2}{20}$$

Consider lamps N, O, Q and S individually,

$$P = \frac{(0.5E)^2}{20}$$

Therefore M and P are brightest, N, O, Q and S are less bright (quarter brightness)

- (b) (i) Using potential divider rule,

$$\left(\frac{1.5}{1.0 + 5.0 + 1.5} \right) 9.0 = V_{AB}$$

$$V_{AB} = 1.8 \, \text{V}$$

- (ii) 0 A in galvanometer: $V_{AD} = 1.5 \, \text{V}$

Using potential divider rule,

$$\frac{1.5}{1.8} = \frac{L_{AD}}{0.800}$$

$$L_{AD} = 0.67 \, \text{m}$$

- (iii) When the $5.0 \, \Omega$ resistor is replaced by a $3.0 \, \Omega$ resistor, the p.d. across wire AB will be larger than $1.8 \, \text{V}$ and hence the balance length of AD will be shorter than in (b)(ii).

8 (a) (i) $a = [0.1 \pm 0.85(1.0 - 0.2)] \alpha = 7.65 - 7.7 \text{ m s}^{-2}$